4 3 2 1

D

C

PRE RELEASE - DO NOT USE
11/30/2022

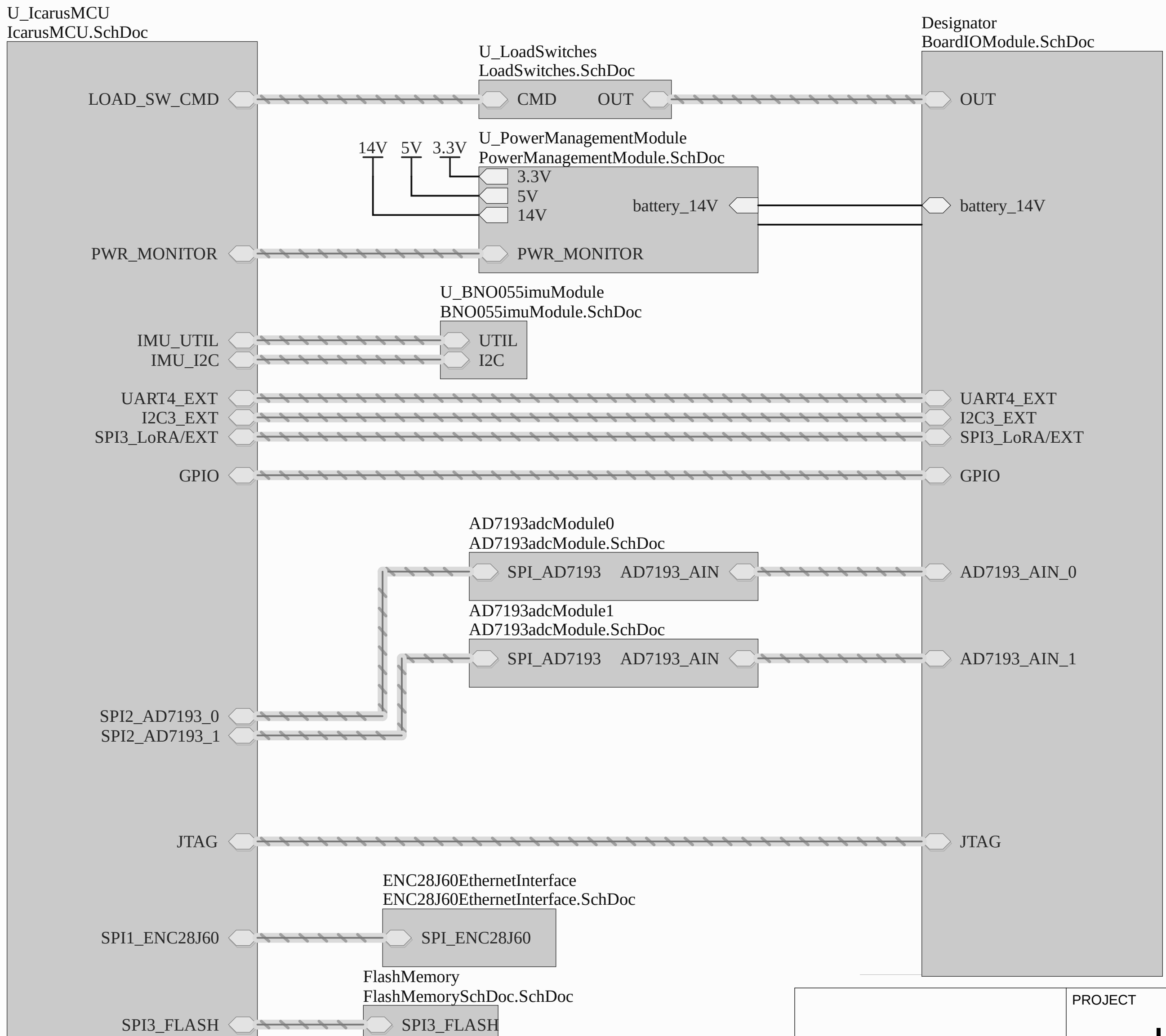
B

4	3	2	1

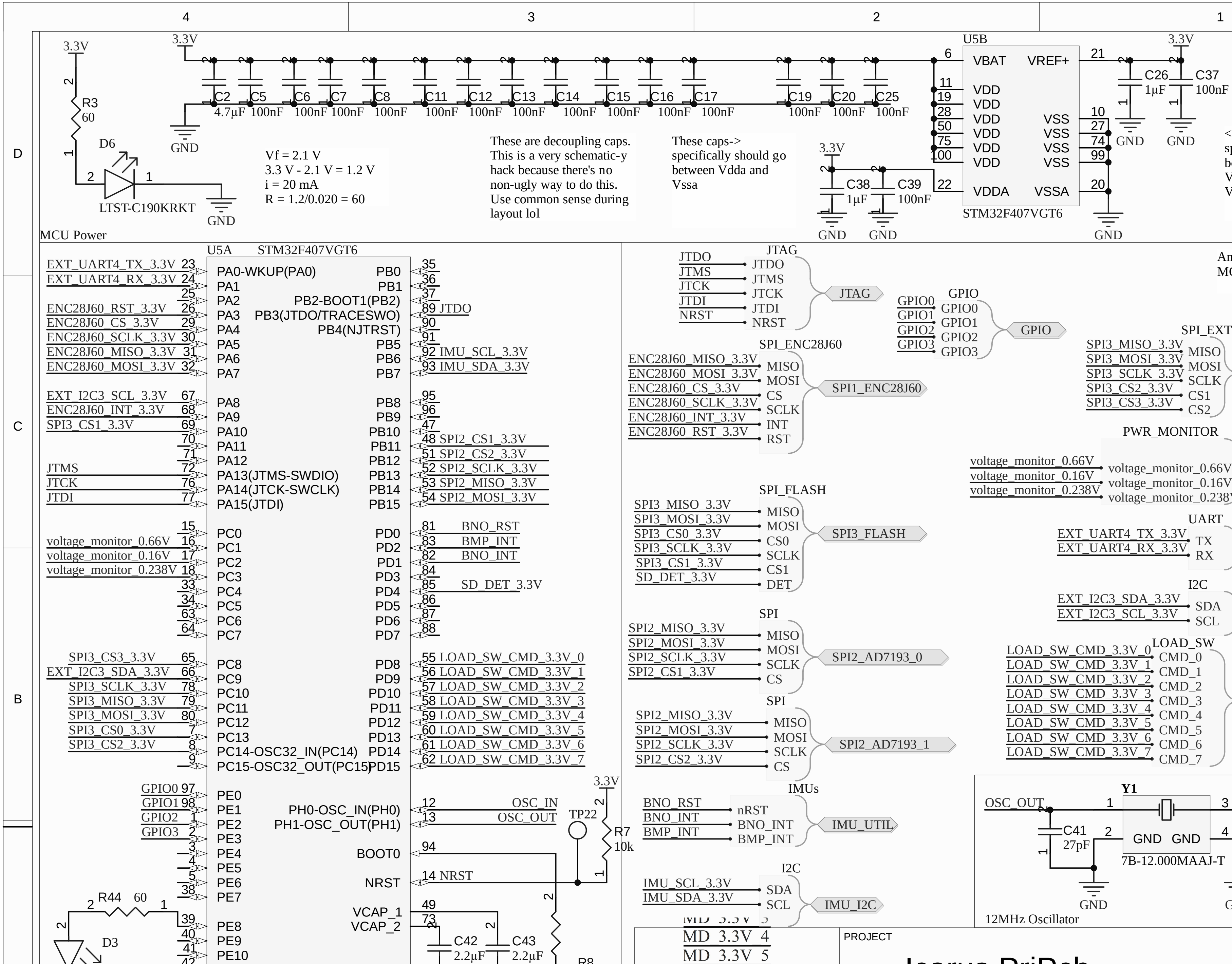
D

C

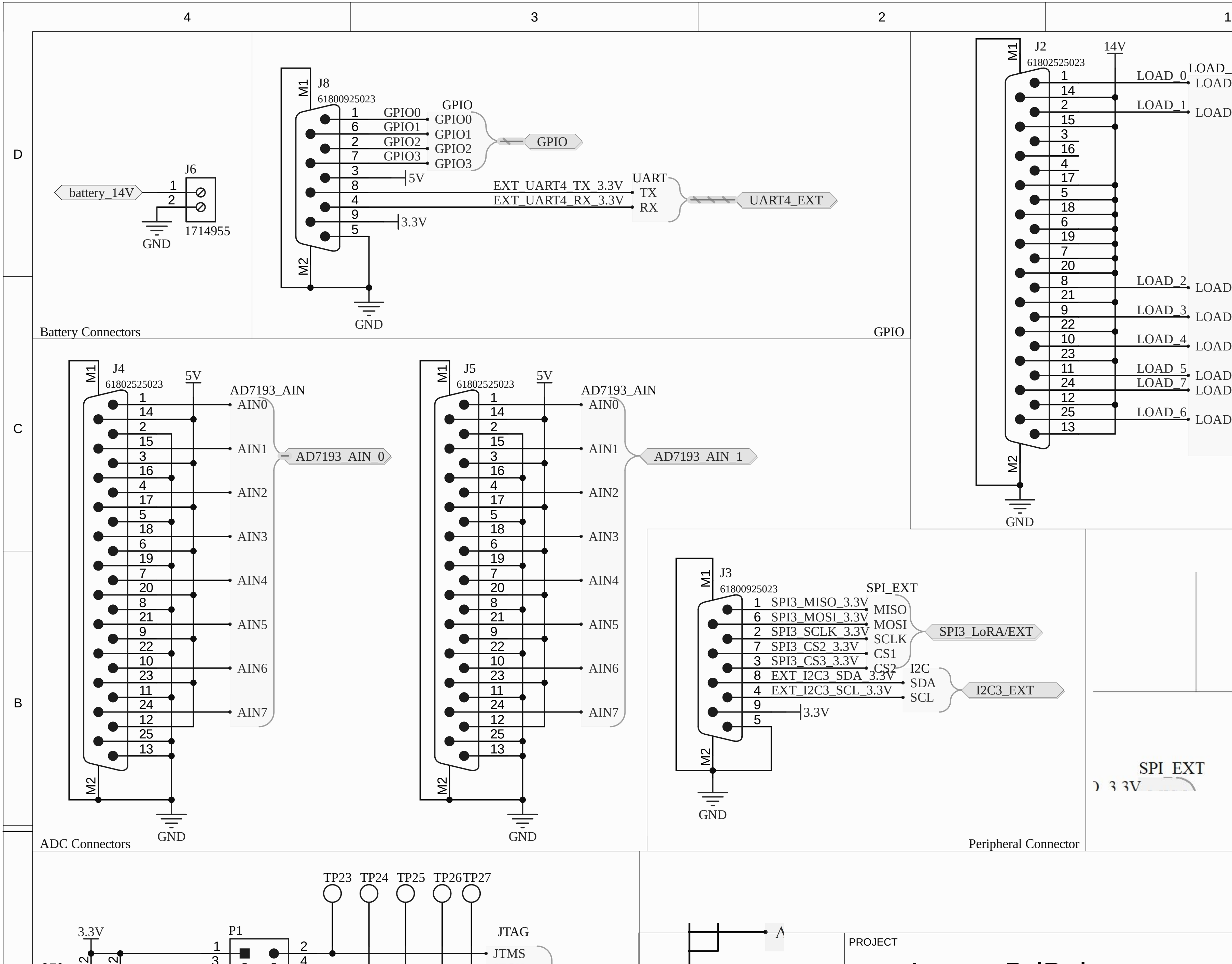
B

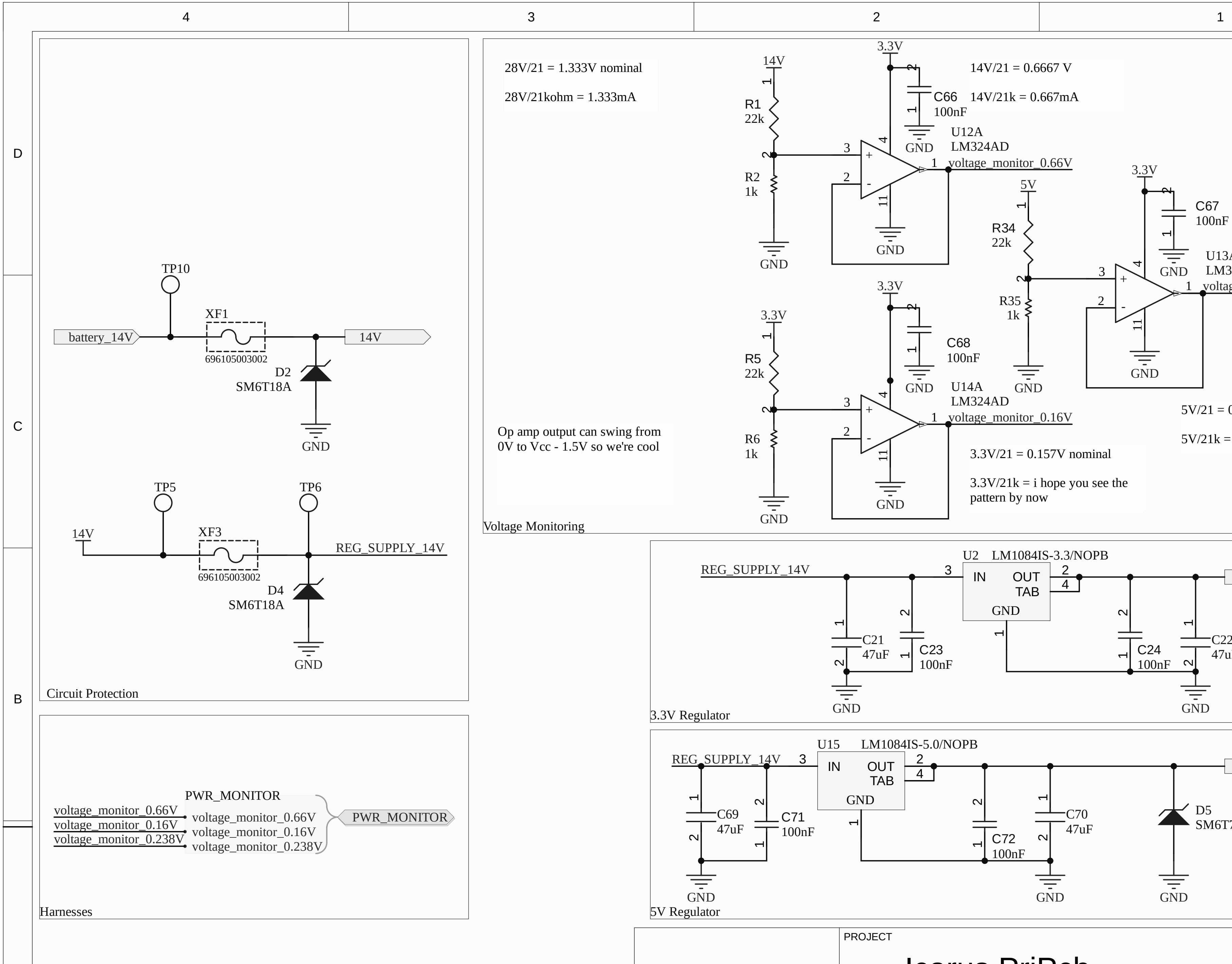


PROJECT



4
3
2
1
D
C
B
3.3V
R3
60
D6
LTST-C190KRKT
GND
C2 4.7µF
C5 100nF
C6 100nF
C7 100nF
C8 100nF
C11 100nF
C12 100nF
C13 100nF
C14 100nF
C15 100nF
C16 100nF
C17 100nF
C19 100nF
C20 100nF
C25 100nF
Vf = 2.1 V
3.3 V - 2.1 V = 1.2 V
i = 20 mA
R = 1.2/0.020 = 60
These are decoupling caps. This is a very schematic-y hack because there's no non-ugly way to do this. Use common sense during layout lol
These caps-> specifically should go between Vdda and Vssa
C38 1µF
C39 100nF
U5B
VBAT
VREF+
VDD
VSS
VDDA
VSSA
STM32F407VGT6
C26 1µF
C37 100nF
MCU Power
U5A STM32F407VGT6
EXT_UART4_TX_3.3V 23 PA0-WKUP(PA0)
EXT_UART4_RX_3.3V 24 PA1
25 PA2
ENC28J60_RST_3.3V 26 PA3
ENC28J60_CS_3.3V 29 PA4
ENC28J60_SCLK_3.3V 30 PA5
ENC28J60_MISO_3.3V 31 PA6
ENC28J60_MOSI_3.3V 32 PA7
EXT_I2C3_SCL_3.3V 67 PA8
ENC28J60_INT_3.3V 68 PA9
SPI3_CS1_3.3V 69 PA10
70 PA11
71 PA12
JTMS 72 PA13(JTMS-SWDIO)
JTCK 76 PA14(JTCK-SWCLK)
JTDI 77 PA15(JTDI)
15 PC0
voltage_monitor_0.66V 16 PC1
voltage_monitor_0.16V 17 PC2
voltage_monitor_0.238V 18 PC3
33 PC4
34 PC5
63 PC6
64 PC7
SPI3_CS3_3.3V 65 PC8
EXT_I2C3_SDA_3.3V 66 PC9
SPI3_SCLK_3.3V 78 PC10
SPI3_MISO_3.3V 79 PC11
SPI3_MOSI_3.3V 80 PC12
SPI3_CS0_3.3V 7 PC13
SPI3_CS2_3.3V 8 PC14-OSC32_IN(PC14)
9 PC15-OSC32_OUT(PC15)
GPIO0 97 PE0
GPIO1 98 PE1
GPIO2 1 PE2
GPIO3 2 PE3
3 PE4
4 PE5
5 PE6
38 PE7
39 PE8
40 PE9
41 PE10
R44 60
D3
PB0 35
PB1 36
PB2-BOOT1(PB2) 37
PB3(JTDO/TRACESWO) 89 JTDO
PB4(NJTRST) 90
PB5 91
PB6 92 IMU_SCL_3.3V
PB7 93 IMU_SDA_3.3V
PB8 95
PB9 96
PB10 47
PB11 48 SPI2_CS1_3.3V
PB12 51 SPI2_CS2_3.3V
PB13 52 SPI2_SCLK_3.3V
PB14 53 SPI2_MISO_3.3V
PB15 54 SPI2_MOSI_3.3V
PD0 81 BNO_RST
PD2 83 BMP_INT
PD1 82 BNO_INT
PD3 84
PD4 85 SD_DET_3.3V
PD5 86
PD6 87
PD7 88
PD8 55 LOAD_SW_CMD_3.3V_0
PD9 56 LOAD_SW_CMD_3.3V_1
PD10 57 LOAD_SW_CMD_3.3V_2
PD11 58 LOAD_SW_CMD_3.3V_3
PD12 59 LOAD_SW_CMD_3.3V_4
PD13 60 LOAD_SW_CMD_3.3V_5
PD14 61 LOAD_SW_CMD_3.3V_6
PD15 62 LOAD_SW_CMD_3.3V_7
PH0-OSC_IN(PH0) 12 OSC_IN
PH1-OSC_OUT(PH1) 13 OSC_OUT
BOOT0 94
NRST 14 NRST
VCAP_1 49
VCAP_2 73
C42 2.2µF
C43 2.2µF
TP22
R7 10k
R8
JTAG: JTDO, JTMS, JTCK, JTDI, NRST
GPIO: GPIO0, GPIO1, GPIO2, GPIO3
SPI_ENC28J60 → SPI1_ENC28J60: ENC28J60_MISO_3.3V MISO, ENC28J60_MOSI_3.3V MOSI, ENC28J60_CS_3.3V CS, ENC28J60_SCLK_3.3V SCLK, ENC28J60_INT_3.3V INT, ENC28J60_RST_3.3V RST
SPI_EXT: SPI3_MISO_3.3V MISO, SPI3_MOSI_3.3V MOSI, SPI3_SCLK_3.3V SCLK, SPI3_CS2_3.3V CS1, SPI3_CS3_3.3V CS2
PWR_MONITOR: voltage_monitor_0.66V, voltage_monitor_0.16V, voltage_monitor_0.238V
SPI_FLASH → SPI3_FLASH: SPI3_MISO_3.3V MISO, SPI3_MOSI_3.3V MOSI, SPI3_CS0_3.3V CS0, SPI3_SCLK_3.3V SCLK, SPI3_CS1_3.3V CS1, SD_DET_3.3V DET
UART: EXT_UART4_TX_3.3V TX, EXT_UART4_RX_3.3V RX
I2C: EXT_I2C3_SDA_3.3V SDA, EXT_I2C3_SCL_3.3V SCL
SPI → SPI2_AD7193_0: SPI2_MISO_3.3V MISO, SPI2_MOSI_3.3V MOSI, SPI2_SCLK_3.3V SCLK, SPI2_CS1_3.3V CS
SPI → SPI2_AD7193_1: SPI2_MISO_3.3V MISO, SPI2_MOSI_3.3V MOSI, SPI2_SCLK_3.3V SCLK, SPI2_CS2_3.3V CS
LOAD_SW: LOAD_SW_CMD_3.3V_0 CMD_0, LOAD_SW_CMD_3.3V_1 CMD_1, LOAD_SW_CMD_3.3V_2 CMD_2, LOAD_SW_CMD_3.3V_3 CMD_3, LOAD_SW_CMD_3.3V_4 CMD_4, LOAD_SW_CMD_3.3V_5 CMD_5, LOAD_SW_CMD_3.3V_6 CMD_6, LOAD_SW_CMD_3.3V_7 CMD_7
IMUs → IMU_UTIL: BNO_RST nRST, BNO_INT BNO_INT, BMP_INT BMP_INT
I2C → IMU_I2C: IMU_SCL_3.3V SDA, IMU_SDA_3.3V SCL
OSC_OUT
Y1
C41 27pF
GND GND
7B-12.000MAAJ-T
12MHz Oscillator
MD_3.3V_4
MD_3.3V_5
PROJECT


4
3
2
1
D
C
B
Battery Connectors
battery_14V
J6
1714955
GND
J8
61800925023
M1
M2
GPIO
GPIO0
GPIO1
GPIO2
GPIO3
5V
3.3V
UART
TX
RX
EXT_UART4_TX_3.3V
EXT_UART4_RX_3.3V
UART4_EXT
GPIO
J2
61802525023
14V
LOAD_0
LOAD_1
LOAD_2
LOAD_3
LOAD_4
LOAD_5
LOAD_7
LOAD_6
ADC Connectors
J4
61802525023
J5
61802525023
5V
AD7193_AIN
AIN0
AIN1
AIN2
AIN3
AIN4
AIN5
AIN6
AIN7
AD7193_AIN_0
AD7193_AIN_1
J3
61800925023
SPI_EXT
SPI3_MISO_3.3V
SPI3_MOSI_3.3V
SPI3_SCLK_3.3V
SPI3_CS2_3.3V
SPI3_CS3_3.3V
EXT_I2C3_SDA_3.3V
EXT_I2C3_SCL_3.3V
MISO
MOSI
SCLK
CS1
CS2
I2C
SDA
SCL
SPI3_LoRA/EXT
I2C3_EXT
Peripheral Connector
SPI_EXT
TP23
TP24
TP25
TP26
TP27
3.3V
P1
C73
JTAG
JTMS
JTCK
PROJECT

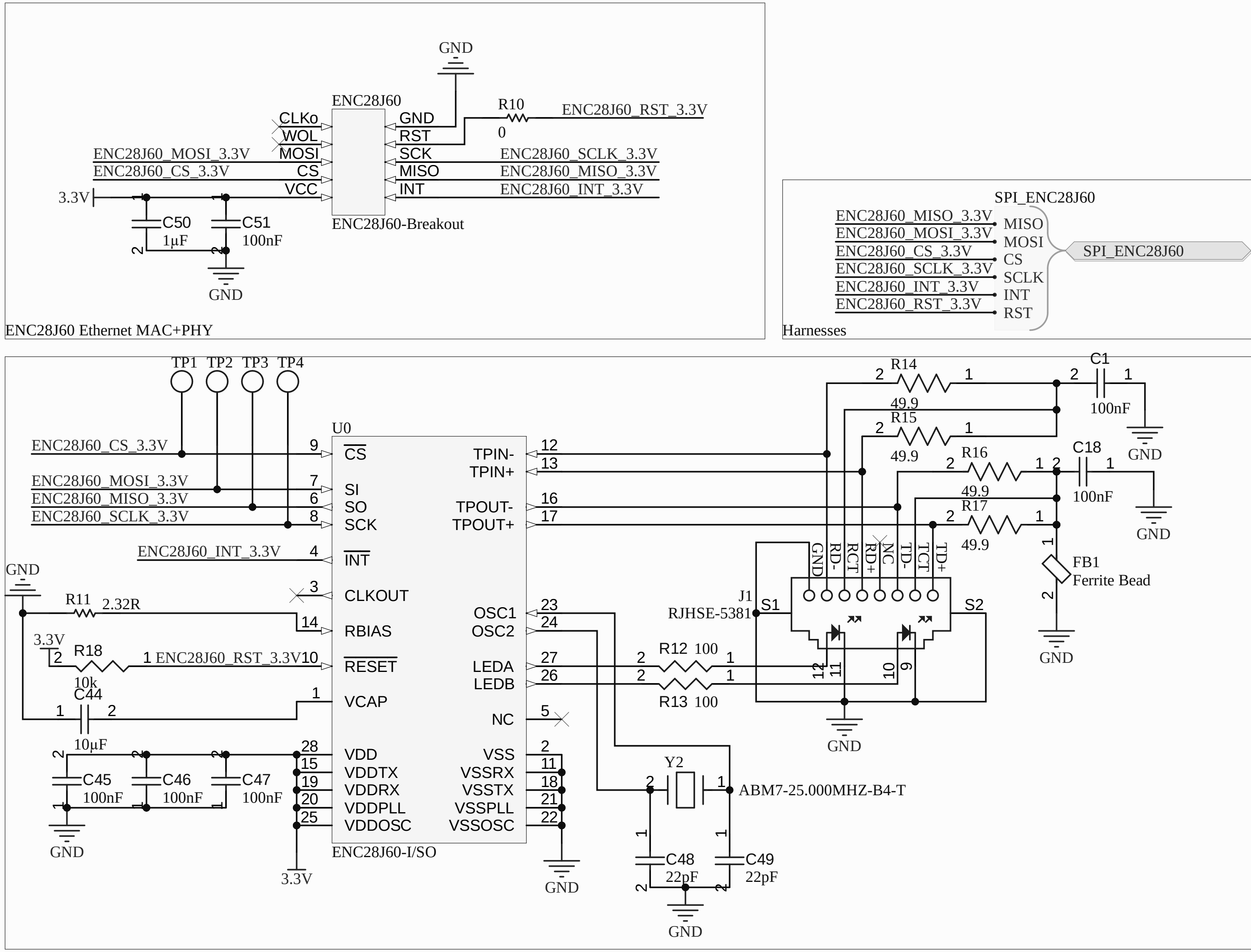

4
3
2
1
D
C
B
28V/21 = 1.333V nominal
28V/21kohm = 1.333mA
14V
R1
22k
R2
1k
GND
3.3V
C66
100nF
U12A
LM324AD
14V/21 = 0.6667 V
14V/21k = 0.667mA
voltage_monitor_0.66V
5V
R34
22k
R35
1k
3.3V
C67
100nF
U13A
LM3
voltag
3.3V
R5
22k
R6
1k
C68
100nF
U14A
LM324AD
voltage_monitor_0.16V
5V/21 =
5V/21k =
Op amp output can swing from
0V to Vcc - 1.5V so we're cool
3.3V/21 = 0.157V nominal
3.3V/21k = i hope you see the
pattern by now
Voltage Monitoring
TP10
XF1
battery_14V
14V
696105003002
D2
SM6T18A
TP5
TP6
XF3
REG_SUPPLY_14V
D4
SM6T18A
Circuit Protection
U2
LM1084IS-3.3/NOPB
IN
OUT
TAB
C21
47uF
C23
100nF
C24
100nF
C22
47u
3.3V Regulator
U15
LM1084IS-5.0/NOPB
C69
47uF
C71
100nF
C72
100nF
C70
47uF
D5
SM6T
5V Regulator
PWR_MONITOR
voltage_monitor_0.66V
voltage_monitor_0.16V
voltage_monitor_0.238V
Harnesses
PROJECT
Icarus PriPcb

4 3 2 1

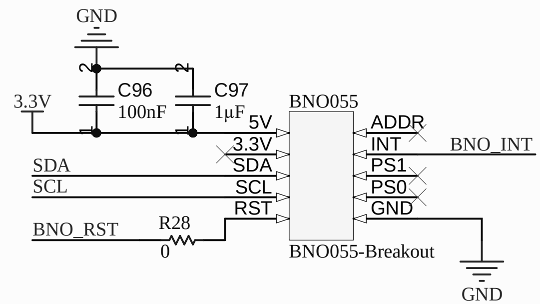
D

C

B



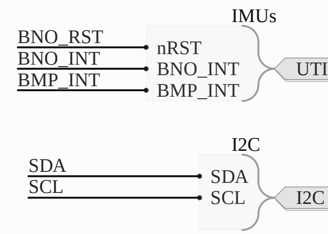
PROJECT



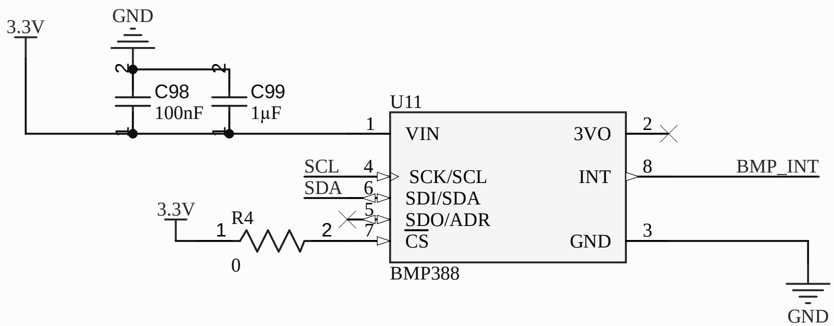
Note: 3.3V is output from onboard reg, 5V is Vin. Vin can take 3.3-5V

PS0 and PS1 are unused as per datasheet

BNO055

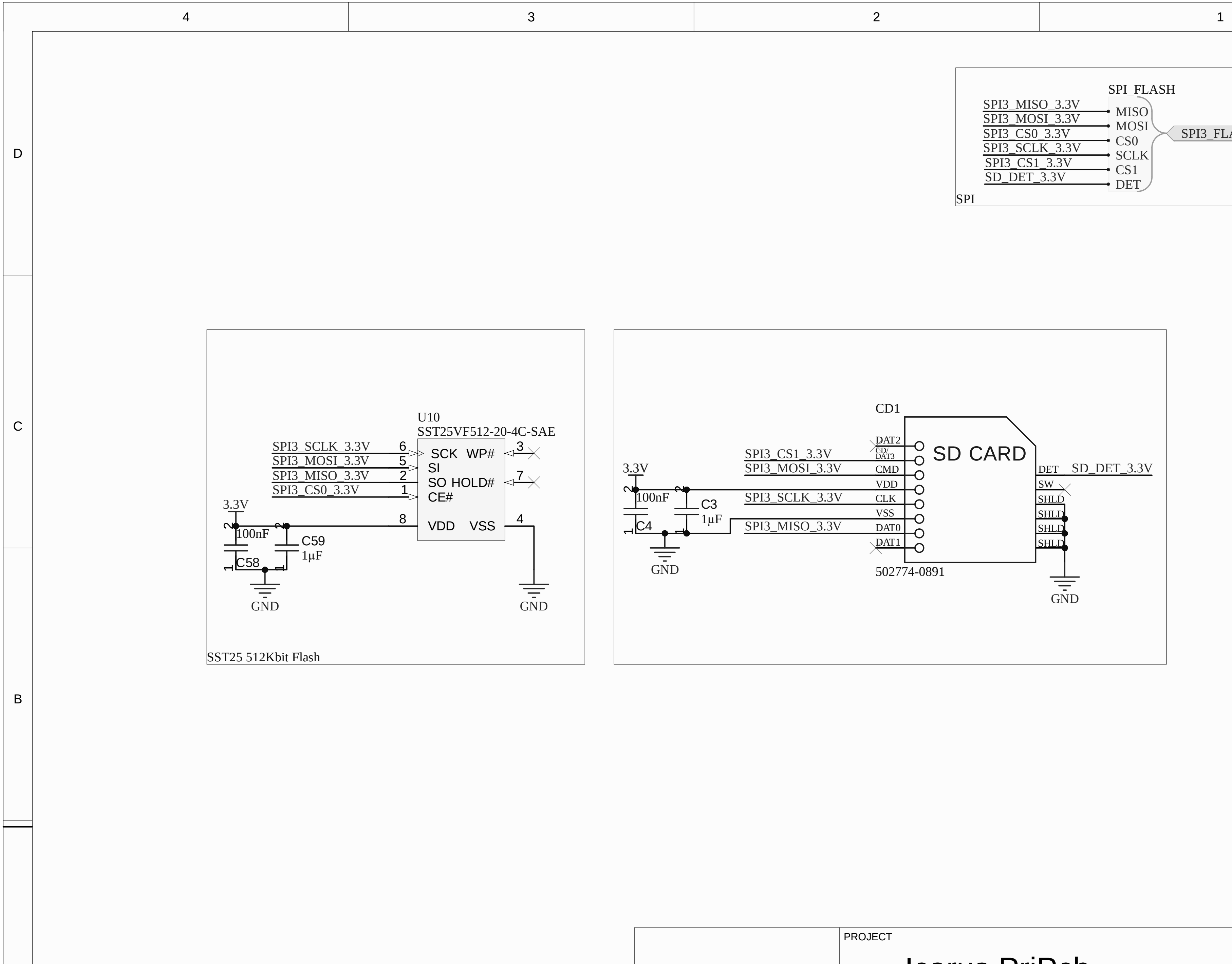


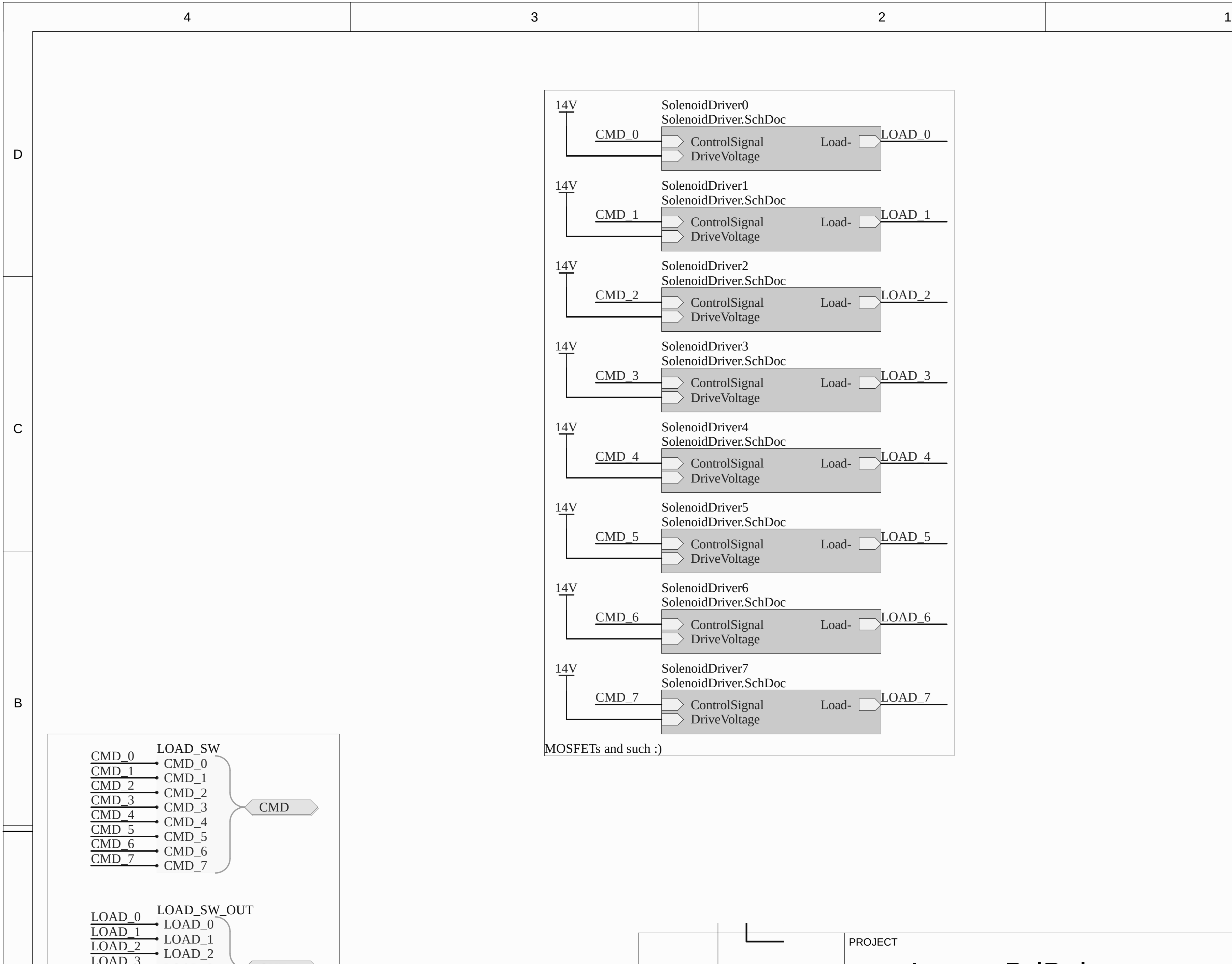
Harnesses



CS has an I/O pin despite being on I2C per the advice of Section 6.3 of the datasheet. 3VO is an internal 3V3 regulator which we don't need.

PROJECT


4
3
2
1
D
C
B
SPI_FLASH
SPI3_MISO_3.3V
SPI3_MOSI_3.3V
SPI3_CS0_3.3V
SPI3_SCLK_3.3V
SPI3_CS1_3.3V
SD_DET_3.3V
MISO
MOSI
CS0
SCLK
CS1
DET
SPI
U10
SST25VF512-20-4C-SAE
SCK
SI
SO
CE#
WP#
HOLD#
VDD
VSS
3.3V
100nF
C58
C59
1µF
GND
SST25 512Kbit Flash
CD1
SD CARD
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SW
SHLD
502774-0891
100nF
C4
C3
1µF
PROJECT



4
3
2
1
D
C
B
14V
SolenoidDriver0
SolenoidDriver.SchDoc
CMD_0
ControlSignal
DriveVoltage
Load-
LOAD_0
14V
SolenoidDriver1
SolenoidDriver.SchDoc
CMD_1
ControlSignal
DriveVoltage
Load-
LOAD_1
14V
SolenoidDriver2
SolenoidDriver.SchDoc
CMD_2
ControlSignal
DriveVoltage
Load-
LOAD_2
14V
SolenoidDriver3
SolenoidDriver.SchDoc
CMD_3
ControlSignal
DriveVoltage
Load-
LOAD_3
14V
SolenoidDriver4
SolenoidDriver.SchDoc
CMD_4
ControlSignal
DriveVoltage
Load-
LOAD_4
14V
SolenoidDriver5
SolenoidDriver.SchDoc
CMD_5
ControlSignal
DriveVoltage
Load-
LOAD_5
14V
SolenoidDriver6
SolenoidDriver.SchDoc
CMD_6
ControlSignal
DriveVoltage
Load-
LOAD_6
14V
SolenoidDriver7
SolenoidDriver.SchDoc
CMD_7
ControlSignal
DriveVoltage
Load-
LOAD_7
MOSFETs and such :)
LOAD_SW
CMD_0
CMD_1
CMD_2
CMD_3
CMD_4
CMD_5
CMD_6
CMD_7
CMD
LOAD_SW_OUT
LOAD_0
LOAD_1
LOAD_2
LOAD_3
PROJECT

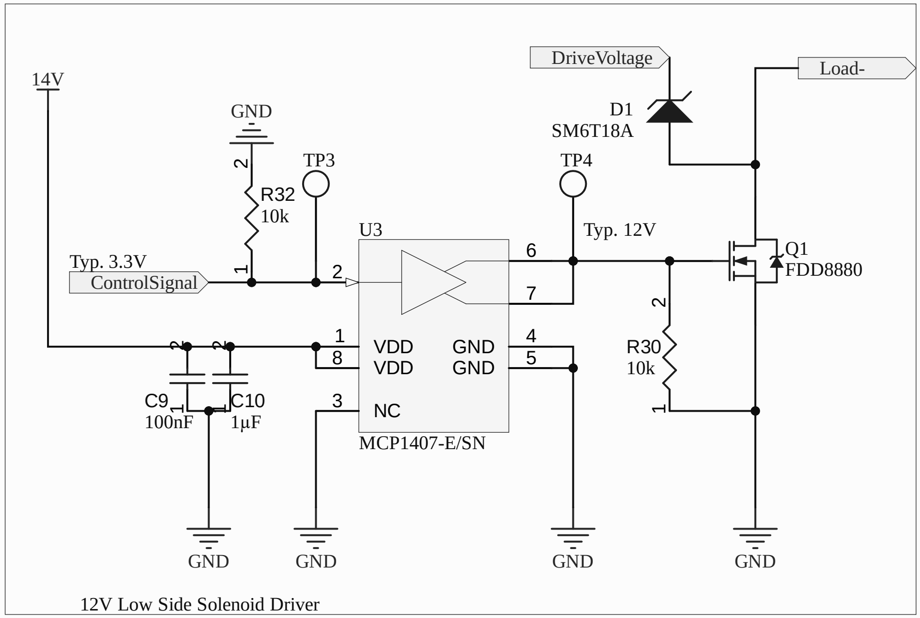
4 3 2 1

D

C

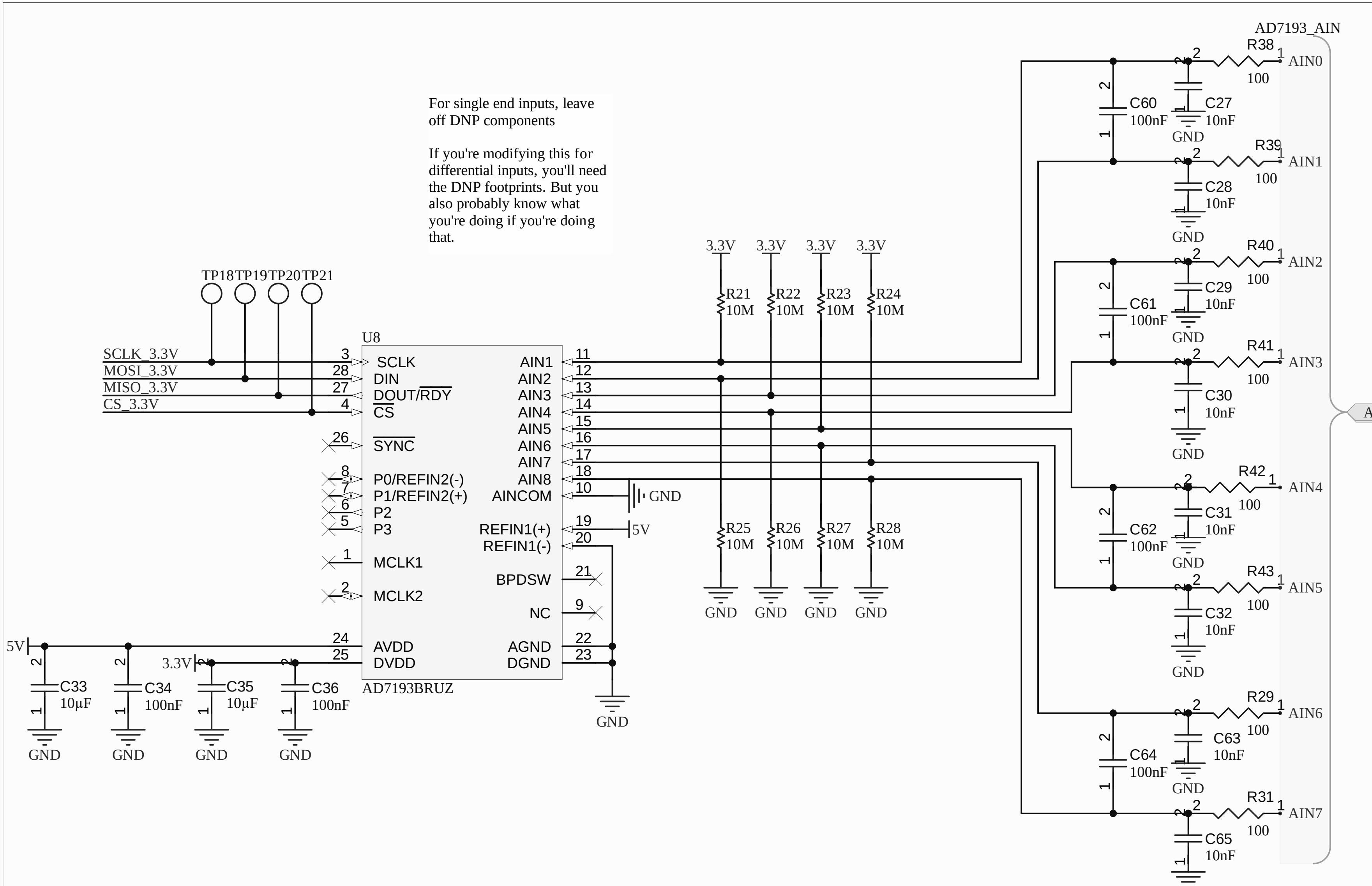
B

MCP1407 max Vdd is 20V, so we use 14V gate drive regardless of solenoid voltage. Max Vgs for FDD8880 is also 20V



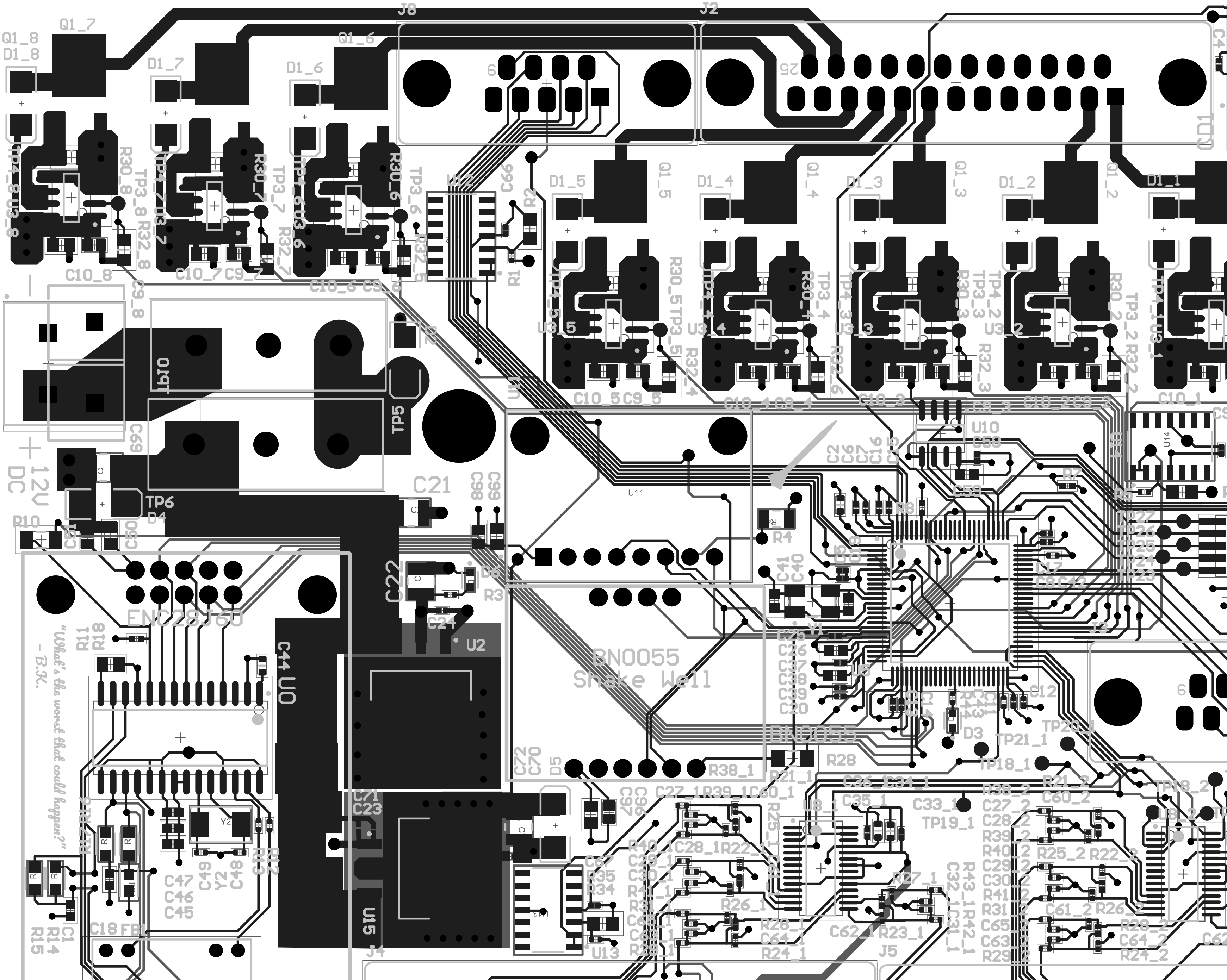
e Solenoid Dri

PROJECT



AD7193 ADC + Bias + Low Pass Filter

PROJECT


J8
J2
BNO055
Shake Well
ENC28J60
+12V DC
"What's the worst that could happen?"
- B.K.
U2
U15
TP5
TP6
C21
C22
C24
C44 U0
U11
U10
U13
J4
J5